Team 044 PianoBot - Summary of Design Changes

MIME Capstone Design II - MusicalBot Encore / PianoBot

Purpose

This document summarizes the major design changes represented by the final Team 044 PianoBot CAD package. It is included as iterative-project documentation for the current design relative to earlier MusicBot/GuitarBot-style concepts, low-fidelity prototypes, and intermediate CAD files.

Major design changes

- **Instrument platform:** The project moved toward a keyboard-based PianoBot that mounts over a Yamaha PSR-530 style 61-key instrument rather than relying on earlier autonomous instrument concepts.
- **Frame and structure:** Early proof-of-concept supports were replaced by a McMaster-Carr 80/20-style 6063 aluminum T-slotted frame. The drawing package includes cut-length drawings for the 1041 mm, 381 mm, and 114 mm frame pieces.
- **Actuation system:** Limited solenoid/key-group prototypes were expanded into a full solenoid bank using 25 N push-pull solenoids arranged into white-key and black-key octave assemblies.
- **Solenoid mounting plates:** Prototype mounting layouts were converted into custom 3D-printed mounting backplates for white-key 7-key, white-key 8-key, and black-key 5-key groups.
- **Electronics mounting:** The final package adds custom support documentation for the MOSFET mount and PCA9685 mount/support to organize control electronics and wiring.
- **Wiring and power distribution:** Terminal blocks, barrier blocks, wiring, power supplies, MOSFET modules, and PCA9685 boards are represented in the assembly drawings and/or the BOM.
- **Documentation consolidation:** The final submission consolidates previously separate Drive/Box/CAD exports into one drawing package PDF, a separate BOM spreadsheet, and this design-change summary.

Final deliverable note

The submitted drawing package contains assembly views, detailed subsystem/part drawings, and custom mount drawings. The separate BOM spreadsheet provides part names, source identifiers, descriptions, material/type notes, quantities, costs, and source links. Off-the-shelf parts are represented through the BOM and assembly views; custom or cut/manufactured parts are represented with detailed drawing sheets where available.